

Youtube Downloader - Developer Documentation (v0.2)

This section provides information for developers who want to understand, modify, or contribute to the YouTube Downloader application.

1. Project Structure

The application consists of the following files:

- **main.py**: The main entry point of the application. Creates the `MainWindow` and starts the Qt event loop.
- **ui/main_window.py**: Defines the `MainWindow` class, which handles the user interface and application logic.
- **ui/download_thread.py**: Defines the `DownloadThread` class, responsible for downloading videos in a separate thread.
- **ui/youtube_search_worker.py**: Defines the `YouTubeSearchWorker` class, which performs YouTube searches in a separate thread.
- **core/downloader.py**: Defines the `VideoDownloader` class, which uses `yt_dlp` to download videos.
- **requirements.txt**: Lists the required Python packages.

2. Dependencies

- **PyQt6**: The GUI framework.
- **requests**: (Not explicitly used in the final code, but often useful for web-related tasks).
- **yt_dlp**: The core library for downloading videos from YouTube and other sites.
- **qtawesome**: Provides Font Awesome icons for the UI.

3. Code Overview

3.1. main.py

- Creates the `QApplication` instance.
- Initializes and shows the `MainWindow`.
- Starts the Qt event loop.

3.2. ui/main_window.py

- **MainWindow Class:**
 - `__init__`: Sets up the UI (creates widgets, layouts, connects signals). Loads user settings.
 - `load_settings`, `save_settings`: Manages user preferences (save path, dark mode).
 - `setup_ui`: Creates all the UI elements.
 - `setup_menu_bar`: Sets up the Help and About menus.
 - `open_help`, `show_about`: Handles actions from the menu bar.
 - `apply_dark_theme`, `apply_light_theme`, `toggle_theme`: Manages the application's theme.
 - `search_videos`: Starts the YouTube search process (creates a `YouTubeSearchWorker`).
 - `display_results`: Displays the search results in the `QListWidget`.
 - `display_error`: Displays search errors.
 - `select_video`: Handles video selection from the list. Populates the quality options.
 - `populate_quality_options`: Populates the quality `QComboBox`.
 - `select_save_path`: Opens a file dialog to choose the download location.
 - `start_download`: Starts the download process (creates a `DownloadThread`).
 - `cancel_download`: Cancels the download.
 - `update_progress`: Updates the progress bar.
 - `download_finished`, `download_error`: Handles download completion or errors.
 - `search_cache`: Caches the search result to avoid re-search with same query.

3.3. ui/youtube_search_worker.py

- **YouTubeSearchWorker Class:**
 - `__init__`: Takes the search query and maximum results as input.
 - `run`: Performs the YouTube search using `yt_dlp`. Emits the `results_signal` with the search results or the `error_signal` if an error occurs.
 - Uses a `ThreadPoolExecutor` for bounded parallelism to improve search speed.

3.4. ui/download_thread.py

- **DownloadThread Class:**
 - `__init__`: Takes the video URL, quality, and save path as input.
 - `run`: Downloads the video using the `VideoDownloader` class. Emits `progress_signal` updates, `finished_signal` on completion, or `error_signal` on failure.

3.5. core/downloader.py

- **VideoDownloader Class:**
 - `download`: Downloads the video using `yt_dlp` with the specified options (quality, output path, etc.).
 - `_progress_hook`: A helper function to handle progress updates from `yt_dlp`.

4. Key Concepts

- **Threading:** `YouTubeSearchWorker` and `DownloadThread` use separate threads to prevent the UI from freezing during long-running operations (searching and downloading).
- **Signals and Slots:** PyQt's signal and slot mechanism is used for communication between threads and the main UI thread. For example, the `results_signal` from `YouTubeSearchWorker` is connected to the `display_results` slot in `MainWindow`.
- **yt_dlp Options:** The code uses various `yt_dlp` options to control the download process:
 - `quiet, no_warnings` : Suppress unnecessary output.
 - `skip_download` : Used during the search to get video information without downloading.
 - `noplaylist` : Ignores playlists.
 - `default_search` : Specifies the search method.
 - `format` : Selects the desired video and audio formats.
 - `merge_output_format` : Ensures the output is in MP4 format.
 - `outtmpl` : Specifies the output file name template.
 - `progress_hooks` : Allows for monitoring the download progress.
- **Caching:** The `search_cache` in `MainWindow` caches search results to improve performance for repeated searches.
- **Settings:** Settings file located in user home directory.

5. GitHub Repository

The source code for this project is available on GitHub: <https://github.com/LongKelvin/yt-downloader>

A pre-built, portable `.exe` version of the application can be found on the releases page: <https://github.com/LongKelvin/yt-downloader/releases/tag/v0.1>

6. Building a Portable Executable (.exe)

If you modify the code and want to create a new `.exe` file, you can use PyInstaller.

1. Install PyInstaller:

```
pip install pyinstaller
```

2. Build the Executable:

Open a terminal in the project directory (where `main.py` is located) and run:

```
pyinstaller --onefile --windowed --name "YouTube Downloader" main.py --collect-all qtawesome
```

- `--onefile` : Creates a single executable file.
- `--windowed` : Prevents a console window from appearing when the `.exe` is run.
- `--name "YouTube Downloader"` : Sets the name of the executable file.
- `main.py` : Specifies the main Python script.
- `--collect-all qtawesome` : Collects all qtawesome files, this is a crucial step.

3. Find the Executable:

After the build process completes, the `.exe` file will be located in the `dist` folder (which PyInstaller creates).

4. Copy user_guide.pdf:

Copy the `user_guide.pdf` to the `dist` folder so it's in same directory with the executable file.

This command creates a single, self-contained executable that includes all the necessary dependencies. Users can run this `.exe` without needing to install Python or any libraries.

Author and Project Information

- **Author:** Long Kelvin
- **Project Name:** yt-downloader
- **GitHub Repository:** <https://github.com/LongKelvin/yt-downloader>
- **Release Date:** March 2025
- **Version:** 0.2
- **Contact:** longkelvin101099@gmail.com